

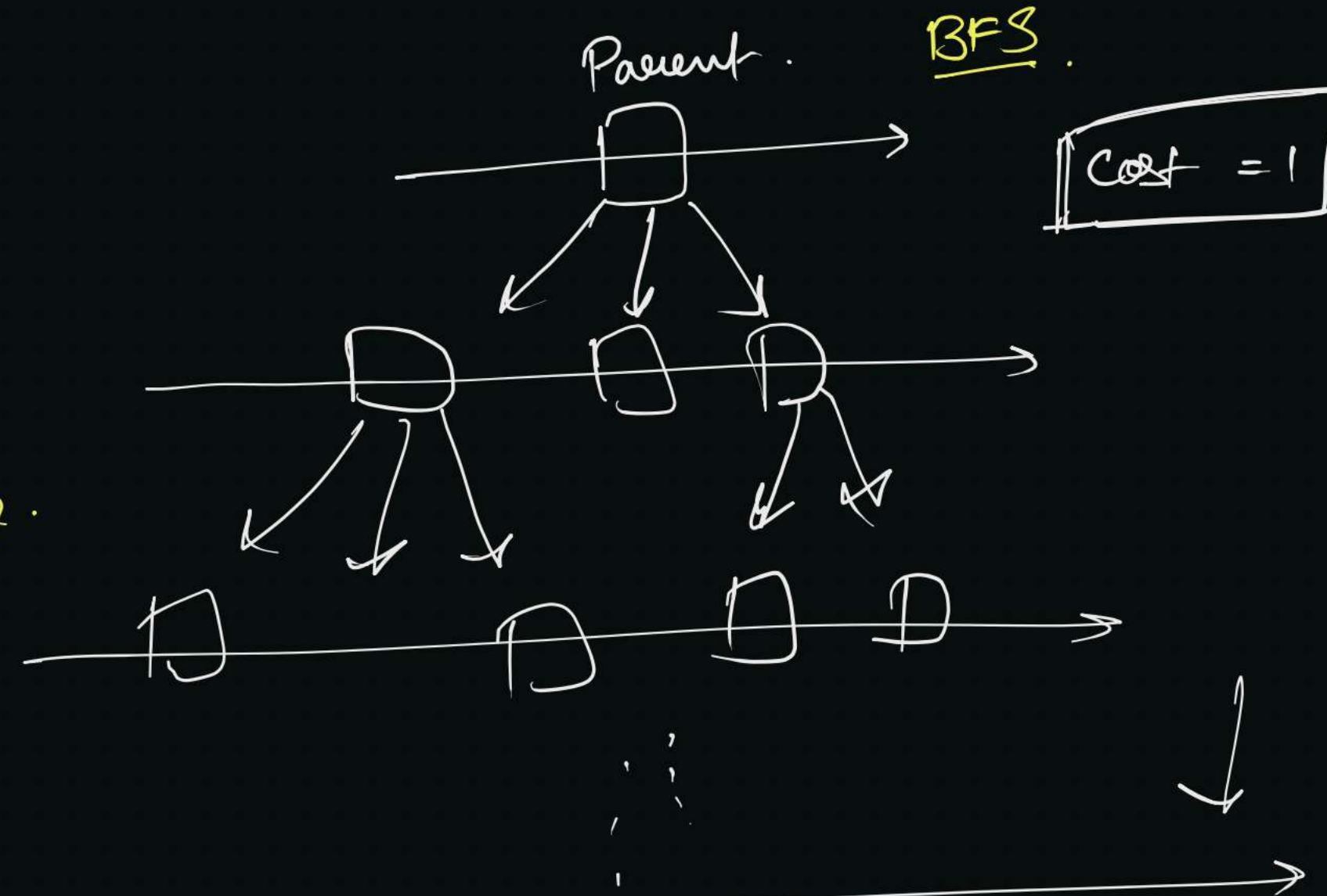
6. Uninformed Search - Uniform cost search

Tuesday, 17 September 2024

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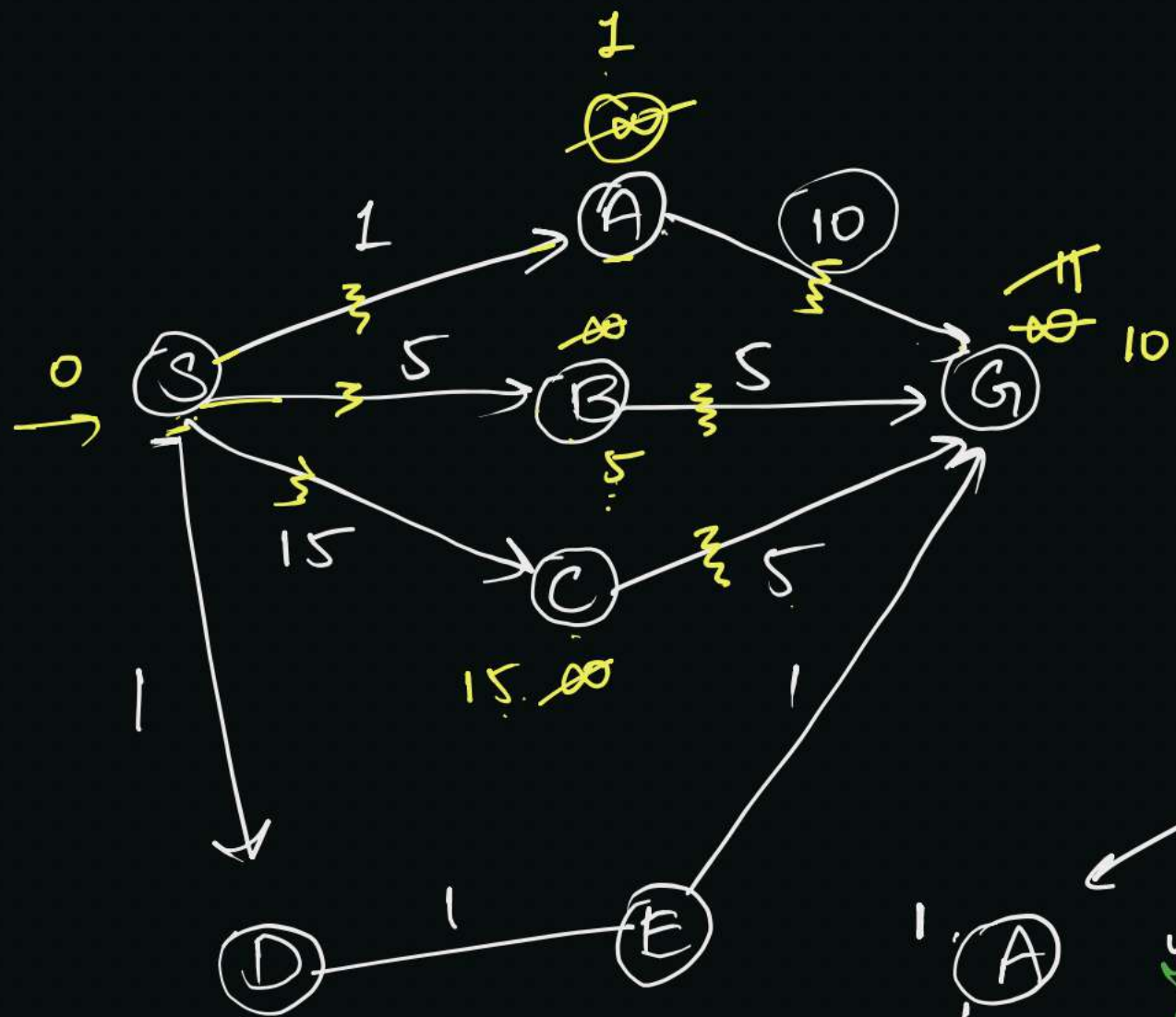
Uniform-Cost Search : (Dijkstra's algorithm)

↳ Data Structure : ?

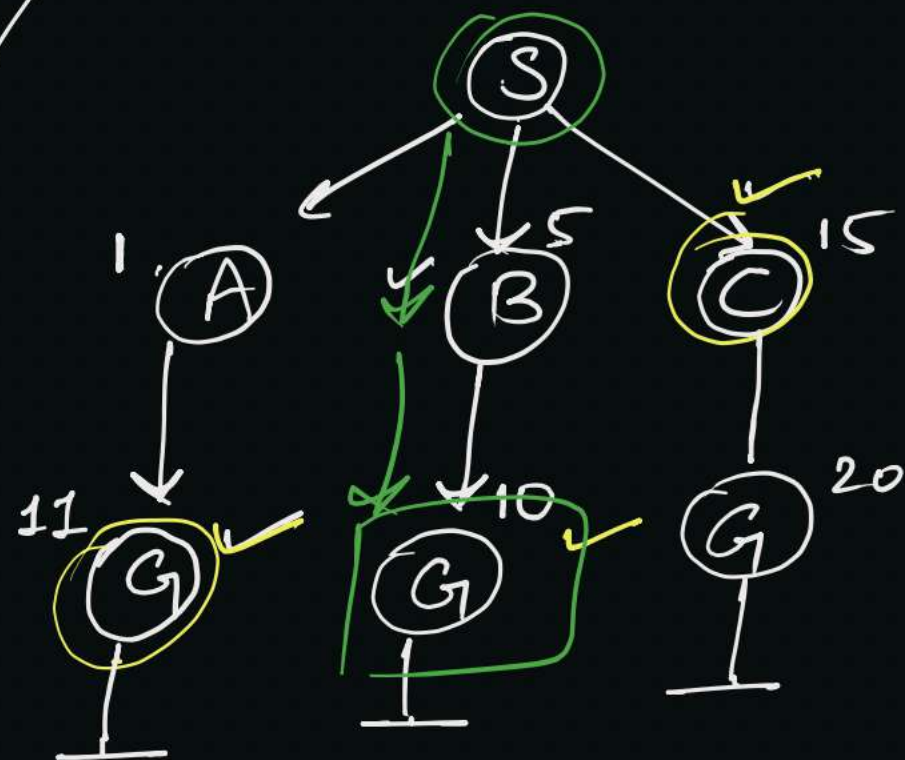


BFS → queue
DFS → stack
UCS → priority queue.

eg.



$$\left. \begin{array}{l} A \rightarrow 1 \\ B \rightarrow 5 \\ C \rightarrow 15 \\ G \rightarrow 10 \end{array} \right\}$$



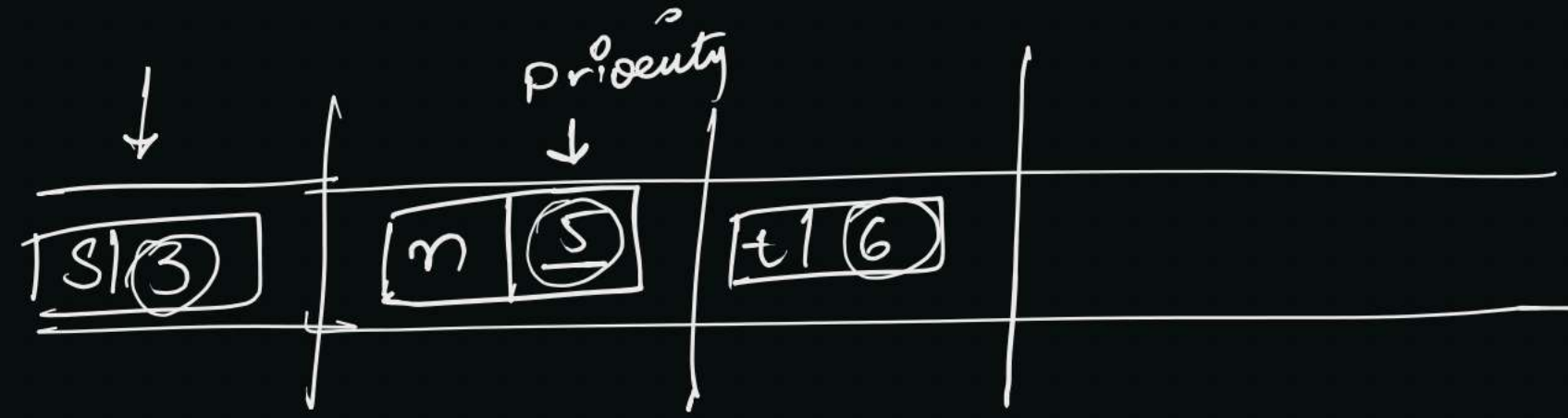
$$S \rightarrow B \rightarrow G$$

pseudo algorithm :-

```

q = priority_queue();
→ q.enqueue(start, 0)
→ while (q is not empty)
{
    → t = q.dequeue();
    if (t == g) → return t;
    else
    {
        for each t' in expand(t, cost_function)
        {
            → if [(t' is not visited) or (t'.cost < visited[t'].cost)]
            {
                visited ← t';
                q.enqueue(t');
            }
        }
    }
}

```

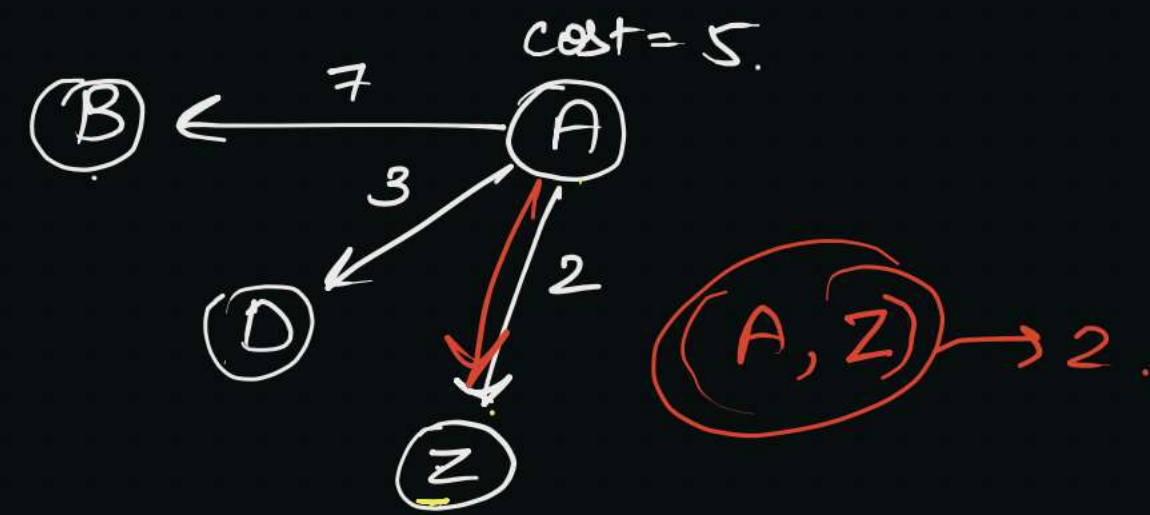


visited.

name of the node.					
A	B	C	D	...	G
0	0	0	0		0

min cost at any state.

12 8



for each t' in expand(t, cost_function)

if [(t' is not visited) or (t'.cost < visited[t'].cost)]

Current cost of t' getting from expand().

previous best cost.

visited ← t';

q.enqueue(t');

Can there be a scenario where we reached the goal & that was not the optimal cost?

(Z, 7), (D, 8), (B, 12)


```

}
}
expand(node, cost-function)
{

```

nodes = []

for all neighbours n' of node

{

$n'.cost = node.cost$ $\xrightarrow{\text{cost of parent}}$

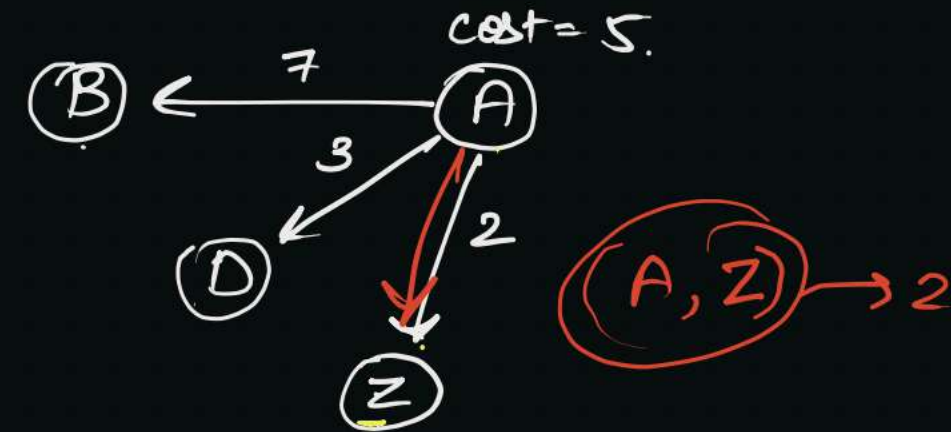
nodes.append(n');

}

return nodes;

}

$expand(A, f) \rightarrow [(Z, 7), (D, 8), (B, 12)]$

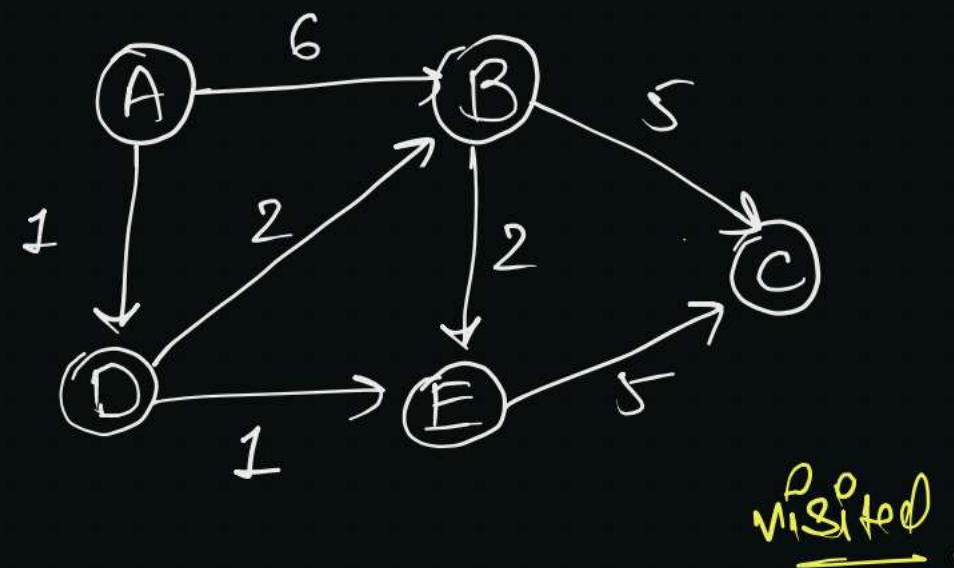


$(Z, 7), (D, 8), (B, 12)$

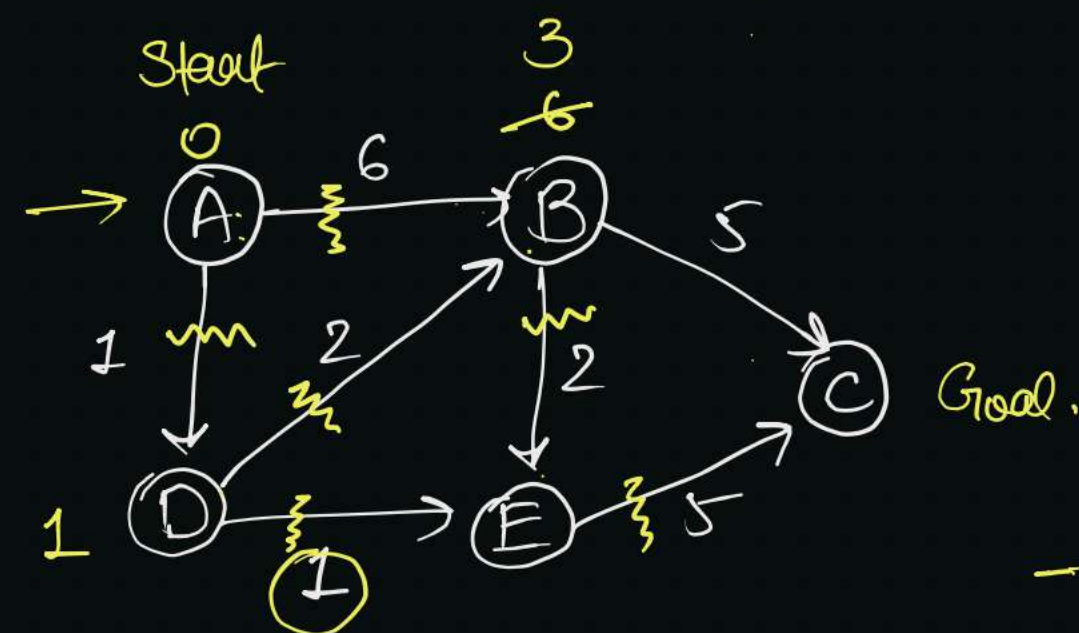
$\xrightarrow{\text{parent}}$ $\xrightarrow{\text{child}}$

$\xrightarrow{\text{cost of that edge. (node} \rightarrow n')$

eg.



A	B	D	E	C		
0	6	1	2	7		
	6	1	2	7		



(A, 0)	(D, 1)	(E, 2)	(B, 3)	(C, 7)
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$$t = (A, 0)$$

$$\text{expand } (A, cf) \leftarrow [(B, 6), (D, 1)]$$

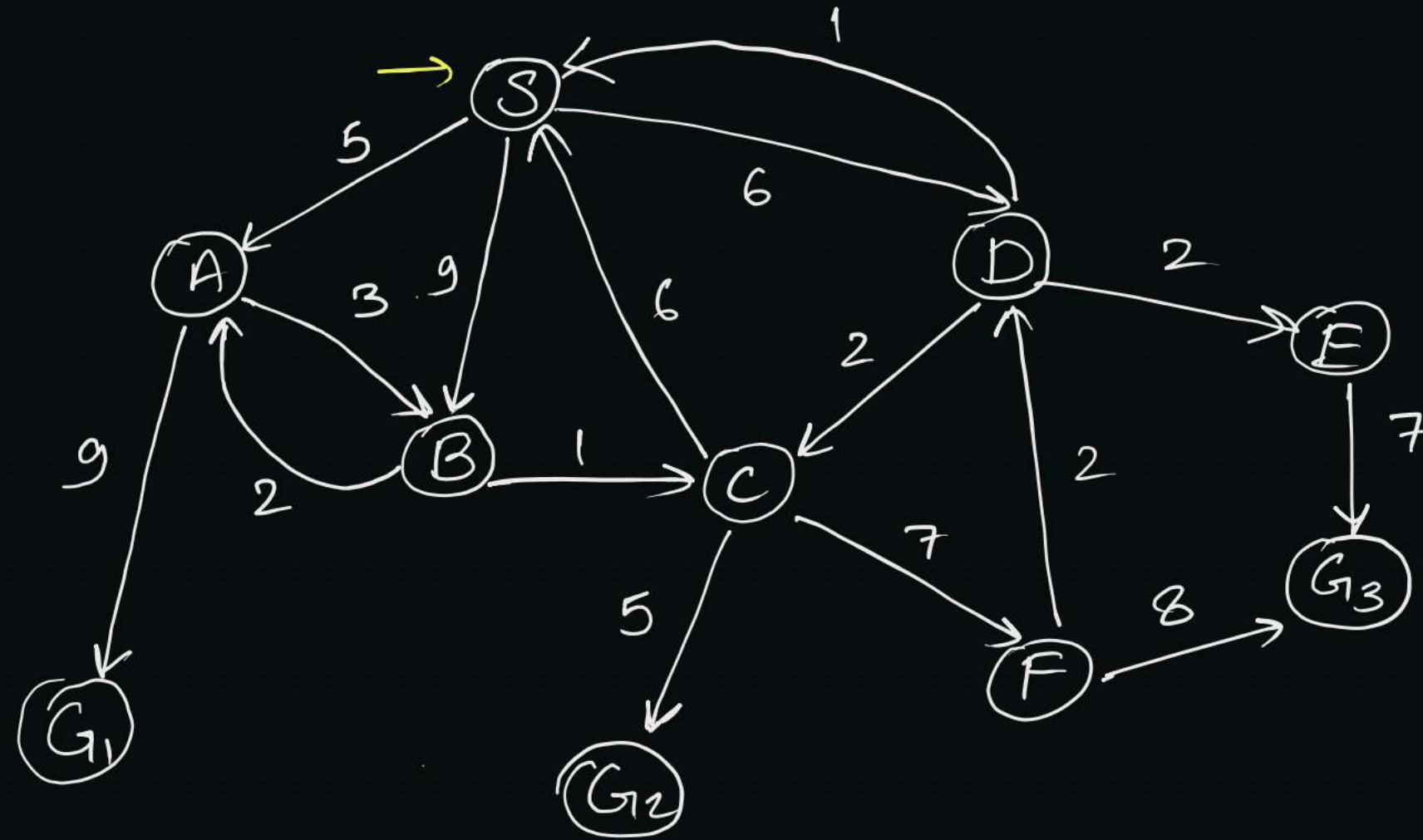
$$t = (D, 1) \rightarrow t' = [(E, 2), (B, 3)]$$

$$t = (E, 2) \rightarrow t' = [(C, 7)]$$

$$t = (B, 3) \rightarrow t' = [(E, 5), (C, 8)]$$

$$t = (C, 7) \Leftarrow \text{we found the goal.}$$

eg.

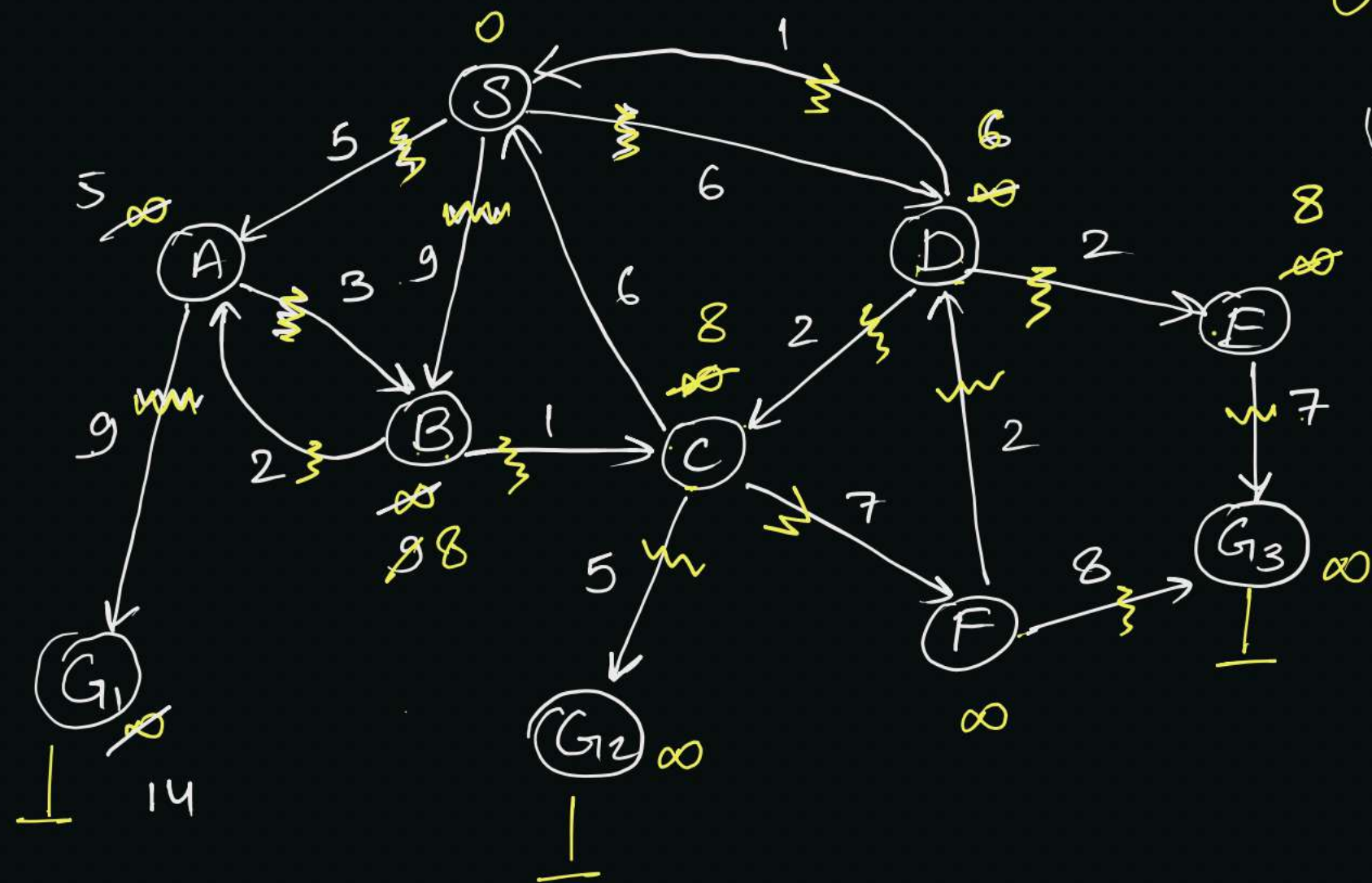


Start = {S}

Goal = {G₁, G₂, G₃}

~~(S, 0)~~, ~~(A, 5)~~, ~~(B, 9)~~, ~~(D, 6)~~, ~~(B, 8)~~, ~~(G₁, 14)~~, ~~(C, 8)~~, ~~(E, 8)~~, (G₂, 13), ~~(F, 15)~~, ~~(G₃, 15)~~

S	A	B	C	D	E	F	G ₁	G ₂	G ₃
0	5 5	9 8	8	6	8	15	14	13	15

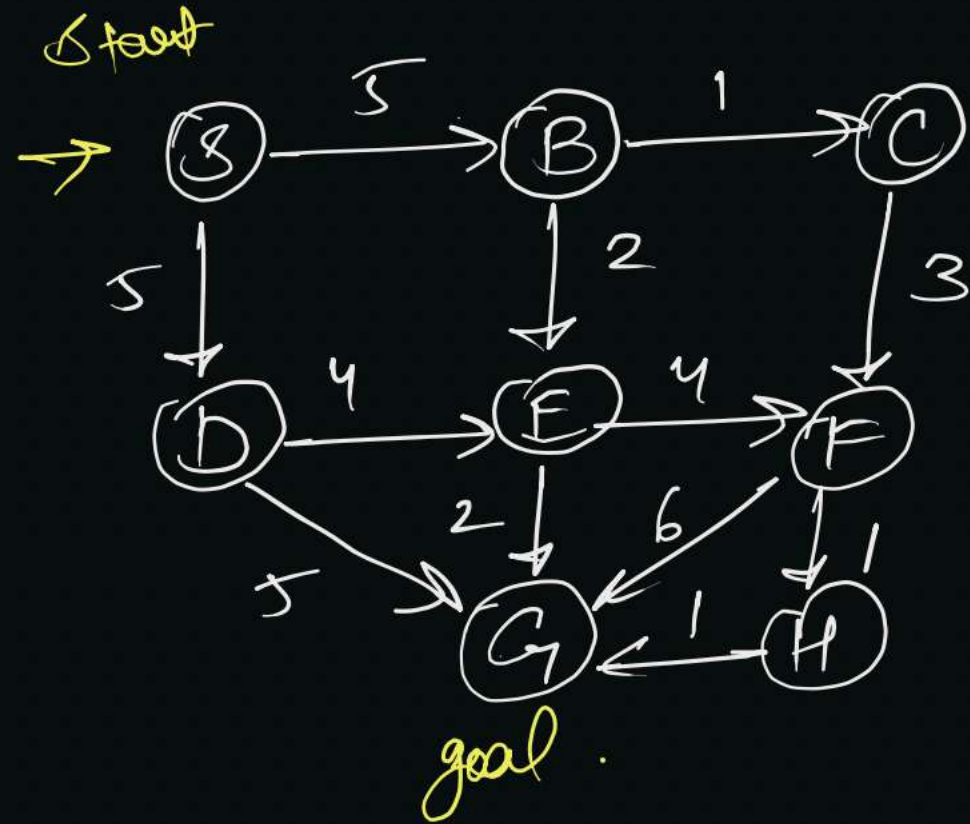


$t = [G_2, 13]$

$t' = [-]$

H.W.

①.



② → Start -

